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Theory Assignment - 2

Section - A

Q. (1) The Entity Relationship Model visually represents how entities relate in a database, offensive a clear blueprint of the data's organisations structure.

(2) In the relation model, data is organised into tables with relationships b/w entities defined through foreign key constraints.

(3) The object - oriented model treat data as a collection of objects, each with unique attribute & method, mirroring real - world entities & interactions.

(4) Integrity constraints ensure the reliability of data by enforcing rules that must be adhesed to maintaining the accuracy of information.

Section B

Q2. Health Care system, discuss data Abstraction and simplify the interactions

Sol^n: Data abstraction is the process of hiding complex database details and showing only relevant information to users.
In a healthcare system data abstraction helps doctors, nurses and staff easily access patient information without dealing with complex database structures.

Eg:- → Doctors can view patient medical history, diagnosis & prescription without seeing how the data is stored in the database.

→ Nurses can update patient vitals & treatment details.

→ Receptionists can easily access appointment & basic patient details.

Through abstraction level:

(i) Physical level : Shows how data is stored inthe database.

(ii) Logical Level: Shows the structure of the database like table & relationships.

(iii) View level: Shows only the table data required by specified users. This makes the system easier to use, secure of efficient for healthcare proffessionals.

Q3. Benefits & challanges of data abstraction.

Ans. Benefits:

(i) Simplicity: Healthcare staff can access required data without understanding database complexity.

(ii) Security: Sensitive patient information can the restricted to authorized users only.

(iii) Data Integrity: Maintains accuracy & consistency of patient data.

(iv) Efficiency: Faster access to patient records improves medical decision making.

Challanges:

(i) Implementation Complexity: Designing multiple abstraction level can be difficult.

(ii) Performance overhead: Additional abstraction layers may slightly affect system performance.

(iii) Maintance Difficult: Updating the system require carefree management of different views & structures.

Section C (Simulation Based)

Q4. Design a database for a university, with entities students, course & professor, using illustration in ER.

Also define the relationships between them.

Sol^n : Entities & Attributes :

(i) Student

Attributes

- Student ID (PK)
- Name
- Age
- Email
- Department

(ii) Course

Attributes

- Curse-ID (PK)
- Course-Name
- Credits
- Departments

(iii) Proffessor

Attribute :

- Proffessor-ID (PK)
- Name
- Email
- Department

Relationship :-

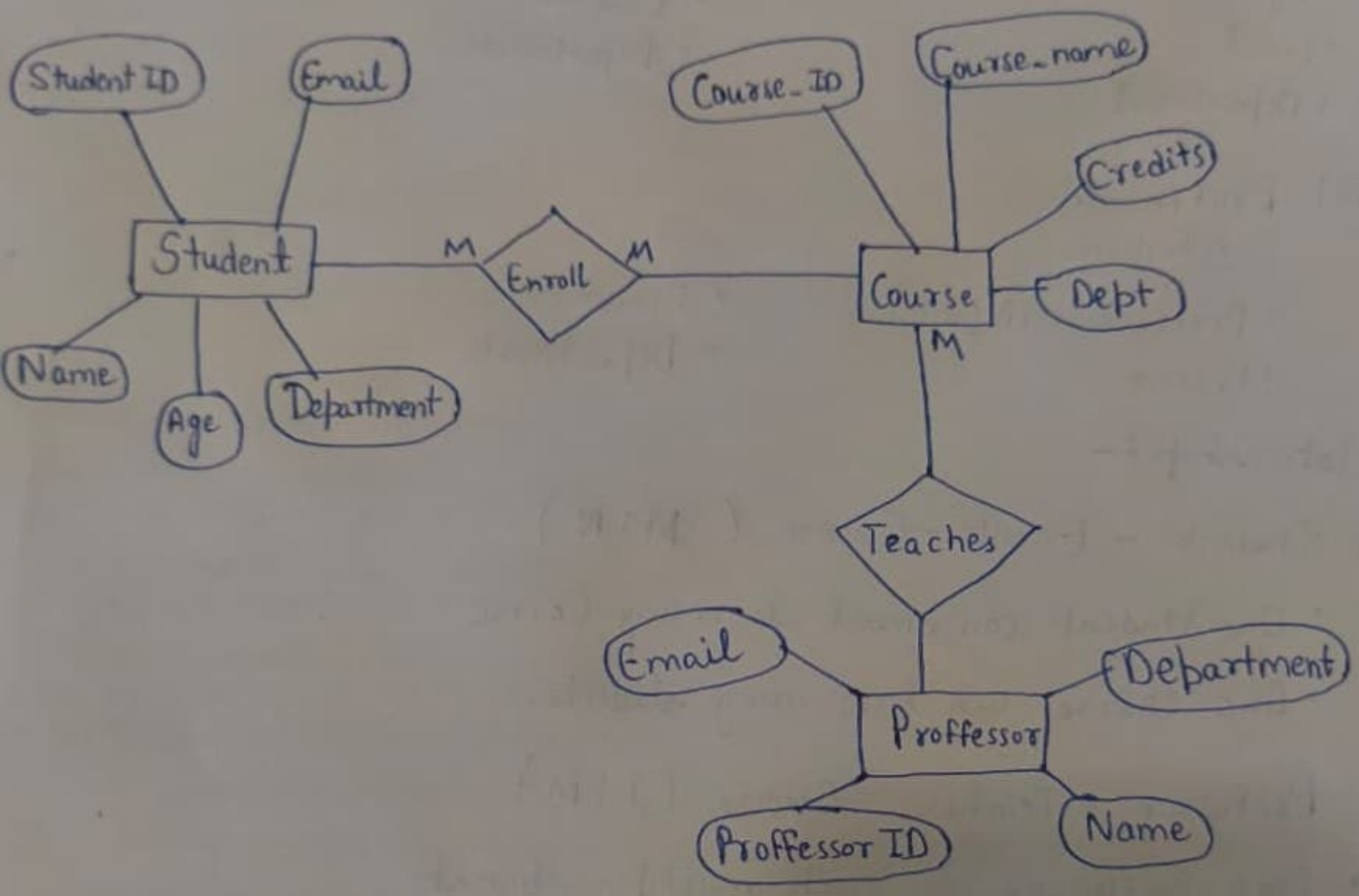
(i) Student - Enrolls - Course (M : N)

- One Student can email in many Course
- One Course can have many students.

(ii) Professor - Teacher - Course (1 : M)

- One Proffessor can teach multiple students
- Each course is taught by one Proffessor

Diagram:


Student ID
Email
Student
Name
Age
Department
M
Enroll
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Course_ID
Course_name
Credits
Course
Dept
M
Teaches
Email
Department
Proffessor
Proffessor ID
Name